The Mystery Motor — speaker script

ANN in fault detection & diagnosis of induction motors · Mhamad Naser Dine & Ali Srour · Supervisor: Prof. Hadi Y. Kanaan

Part	Slides	Speaker	Time
Problem, idea, fingerprints, features	1 – 6	Mhamad	3:05
Network, data, point A, hidden layer, results, reliability, conclusion	7 – 14, 16	Ali	3:50
Total talk			**6:55**
Live demo (before the conclusion)	15	Mhamad	2:00

Green bold = the words written on the slide (you can read them from the screen) · CLICK = press → (an element appears) · speak slowly, about 130 words per minute.

1 This motor looks healthy. Is it?

Mhamad · 0:00 – 0:20

Good morning. I am Mhamad Naser Dine, and this is Ali Srour.

Look at this motor and its current: a clean sine wave.

This motor looks healthy. Is it?

Keep this question in mind. We will answer it live at the end.

2 Motors break slowly, so we want to catch the fault early

Mhamad · 0:20 – 0:55

Induction motors run pumps, fans and conveyors in every factory. When one stops, the production stops. And **motors break slowly**.

Most faults come from three places. CLICK **Rotor bars** crack and break. CLICK **Stator windings**: the insulation fails and turns short-circuit. CLICK **Bearings** wear out.

These faults start small and grow over weeks.

CLICK So our goal: **detect the fault while it is still small.**

3 A broken rotor bar is almost invisible in the current

Mhamad · 0:55 – 1:20

Here we broke **3 rotor bars**: the orange bar.

Teal is the current of the **healthy motor**. Dashed orange is the motor with **3 broken bars**.

CLICK **The two currents are almost identical: the eye cannot see the fault.**

4 …but it is clearly visible in the frequency spectrum

Mhamad 1:20 – 2:00

But it is clearly visible in the frequency spectrum.

This video shows the same motor while the fault grows.

Healthy motor: one peak at 50 hertz.

With **1 broken bar**, two side peaks appear, at (1 ± 2s)·f, exactly where the theory says.

With **3 broken bars**, they get even stronger.

The fault grows, and its fingerprint grows with it. This is called motor current signature analysis.

5 Each fault leaves a different fingerprint

Mhamad 2:00 – 2:35

Each fault leaves a different fingerprint.

CLICK **Broken bars**: side peaks. CLICK **Bearing defect**: peaks linked to the bearing. CLICK **Inter-turn short**: the **Park circle becomes an ellipse**. CLICK **Supply unbalance**: the voltage itself is unbalanced.

The network must learn all four patterns, plus the healthy one.

6 Each motor becomes 19 numbers

Mhamad 2:35 – 3:05

A network cannot read thousands of samples. So **each motor becomes 19 numbers**: **time**, **balance**, **Park**, **spectrum** and **slip**.

For our faulty motor, the orange **broken-bar sidebands** are the highest.

Ali will now show how the network learns.

7 A neural network learns these fingerprints

Ali 3:05 – 3:40

Thank you, Mhamad. The 19 numbers become the **19 inputs** of a neural network, with **15 hidden** neurons and **5 outputs**, one per condition.

CLICK The hidden layer combines the inputs. CLICK The highest output is the diagnosis: **Broken bars, 96 %**.

It learns with **back-propagation**: the error goes backward and corrects every weight. After many examples, its error becomes very small.

8 The network learns from 1500 virtual motors

Ali 3:40 – 3:55

To learn, it needs examples. **The network learns from 1500 virtual motors**, **300** per condition, with different loads and noise. **One dot = 10 simulated motors.**

9 We split them: learn, know when to stop, final exam

Ali 3:55 – 4:25

We split them, exactly as in the course.

Training, 60 %: it **adjusts the weights**.

Test, 20 %: it **stops at point A**.

Validation, 20 %: never seen during learning. It is **the final, honest grade**. Every number we show comes from this set.

10 It stops at point A, before it starts memorising

Ali 4:25 – 4:50

CLICK The **training error keeps falling**.

CLICK But the **test error stops falling at A**, epoch 149.

CLICK **Point A**: **we keep these weights**, before the network starts memorising.

11 A single layer can only draw a straight line…

Ali 4:50 – 5:10

Why do we need a hidden layer? Here we keep only two features.

Healthy motors sit in one corner. **A single layer can only draw a straight line**: **Perceptron, 86.7 %**.

12 …a hidden layer bends the boundary

Ali 5:10 – 5:25

A hidden layer bends the boundary around the corner: **92.5 %**.

That is why we use a multilayer perceptron.

13 MLP-BP is the best: 94 %, the RBF only 78 %

Ali 5:25 – 6:00

We compared seven networks, all written from scratch, each trained ten times.

MLP-BP is the best: 94 %. Momentum and adaptive η are almost the same.

MLP-LM is the fastest. **ADALINE** gets 90 %, and **the RBF only 78 %**: its Gaussian neurons only see a small region, so they cannot cover 19 dimensions.

14 Even one broken bar is found 89 % of the time

Ali 6:00 – 6:30

We tested on **500 new motors never seen**. **Even one broken bar is found 89 % of the time.**

CLICK **100 %** of medium and severe faults. CLICK **8 hidden neurons are enough.**

CLICK **Our limit**: at **5 % sensor noise**, accuracy drops to **31 %**.

Now, back to our mystery motor. Mhamad.

15 Live demo: is the mystery motor healthy?

Mhamad · DEMO · 2:00

0:00 Back to our question: **is the mystery motor healthy?** CLICK **The program creates a motor and hides its condition.** CLICK **The network reads only the current.** CLICK **We reveal the truth.**

0:15 Switch to the app (page “Live diagnosis demo”, Saved networks, MLP-BP, noise 1 %). Choose “Random unknown motor” → click “Acquire signals & diagnose”. Say: “Even we do not know the condition.”

0:30 Read the verdict aloud, e.g. “Faulty — broken rotor bars, 96 %.” Say: “The bars show how sure it is.”

0:45 Click “Reveal the true condition”. Say: “The network was right.” (If wrong: “This is a small fault at light load — the hard case.”)

1:00 Do one more random motor: click, read, reveal.

1:20 Choose the condition yourself: Broken rotor bars, severity 1, load 40 %. Click. Say: “Even one broken bar is detected. Look at the small side peaks.”

1:50 Say: “From the current to the diagnosis in one second. Ali will conclude.” → back to the slides.

16 From the current to the diagnosis

Ali · 6:30 – 6:55

To conclude: **from the current to the diagnosis**.

94 % with MLP-BP, **89 % of early faults found**, and **an honest evaluation**: 3 sets, point A, 10 runs.

Next: real motor data, the severity, and the remaining life.

Thank you — questions?

Before the presentation

- Rehearse twice with a timer; the talk should end around 7 minutes.
- Open the app on “Live diagnosis demo” (Saved, MLP-BP, noise 1 %) and run it once before you start.
- Keep a backup screen recording of the demo in case the app fails.
- Switch speaker smoothly: “Ali will now show…” / “Back to our mystery motor. Mhamad.”